CO3-AT1 (CASE STUDY)

FODS-DSA0405

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1)Case Study-Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

SOLUTION:

Data (₹ lakhs):

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Mean, Median and Mode

Mean

$$\text{Mean} = \frac{4+5+5+6+7+8+8+9+30}{9} = \frac{82}{9} = 9.11$$

Mean = ₹9.11 lakhs

Median
There are 9 observations.
Middle value (5th value):

$$\boxed{7}$$

Median = ₹7 lakhs

Mode
Most frequent values are:
5 (appears twice)
8 (appears twice)
Mode = ₹5 lakhs and ₹8 lakhs (Bimodal)

2. Range

$$30 - 4 = 26$$

Range = ₹26 lakhs

3. Variance and Standard Deviation

Mean = 9.11

Salary	1) $x - \mu$	2) $(x - \mu)^2$
4	3) -5.11	4) 26.12
5	5) -4.11	6) 16.90
5	7) -4.11	8) 16.90
6	9) -3.11	10) 9.68
7	11) -2.11	12) 4.46
8	13) -1.11	14) 1.23
8	15) -1.11	16) 1.23
9	17) -0.11	18) 0.01
30	19) 20.89	20) 436.34

$$\sum (x - \mu)^2 = 512.87$$

Variance

$$\sigma^2 = \frac{512.87}{9} = 56.99$$

Standard deviation

$$\sigma = \sqrt{56.99} = 7.55$$

Variance = 56.99

Standard Deviation = 7.55

4. Outlier

The salary ₹30 lakhs is much larger than all others.

Outlier = ₹30 lakhs

5. Best Measure of Central Tendency

Median (₹7 lakhs) is the best measure because it is not affected by the extreme outlier (₹30 lakhs), whereas the mean is increased significantly.

2) Case Study-Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.

2. Find the range.

3. Calculate the variance and standard deviation.

4. Determine whether the data is likely to be right-skewed or left-skewed.

5. Explain why the median may be more useful than the mean.

SOLUTION:

Data (ms):

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Mean and Median

Sum

$$1505$$

Mean

$$\frac{1505}{10} = 150.5$$

Mean = 150.5 ms

Median

Middle values:

132 and 135

$$\frac{132 + 135}{2} = 133.5$$

Median = 133.5 ms

2. Range

$$300 - 120 = 180$$

Range = 180 ms

3. Variance and Standard Deviation

$$\sum (x - \mu)^2 = 26906.5$$

Variance

$$\frac{26906.5}{10} = 2690.65$$

Standard deviation

$$\sqrt{2690.65} = 51.87$$

Variance = 2690.65

Standard Deviation = 51.87 ms

4. Skewness

The value 300 ms is much larger than the remaining values.

Therefore,

The distribution is Right-Skewed (Positively Skewed).

5. Why Median is Better

The median is not affected by the extreme response time (300 ms).

The mean increases because of the outlier, making the website appear slower than it usually is.

3) CASE STUDY- Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the IQR.
3. Use the 1.5 × IQR rule to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.

SOLUTION:

Marks

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3

There are 12 values.

Q2 (Median)

$$\frac{52+55}{2} = 53.5$$

Lower Half

42,45,48,50,52,52

Q1

$$\frac{48+50}{2} = 49$$

Upper Half

55,58,60,62,65,95

Q3

$$\frac{60+62}{2} = 61$$

Q1 = 49

Q2 = 53.5

Q3 = 61

2. IQR

$$61 - 49 = 12$$

IQR = 12

3. Outlier using 1.5 × IQR

$$1.5 \times 12 = 18$$

Lower limit

$$49 - 18 = 31$$

Upper limit

$$61 + 18 = 79$$

Only

$$95 > 79$$

Outlier = 95

4. Mean and Median

Sum

$$684$$

Mean

$$\frac{684}{12} = 57$$

Mean = 57

Median

$$53.5$$

Median = 53.5

5. Effect of Outlier

The score 95 increases the mean from the typical student performance, while the median remains almost unchanged. Hence, the median better represents the class performance.

4) Case Study - E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Solution:

Orders

18,20,22,22,24,25,26,28,30,35

1. Mean, Median and Mode

Sum

$$250$$

Mean

$$25$$

Median

$$\frac{24+25}{2}=24.5$$

Mode

22 occurs twice.

Mean = 25

Median = 24.5

Mode = 22

2. Variance and Standard Deviation

$$\sum (x-\mu)^2 = 204$$

Variance

$$20.4$$

Standard deviation

$$\sqrt{20.4} = 4.52$$

Variance = 20.4

Standard Deviation = 4.52

3. Coefficient of Variation

$$CV = \frac{4.52}{25} \times 100$$
$$CV = 18.08\%$$

Coefficient of Variation = 18.08%

4. Meaning of Standard Deviation

A standard deviation of 4.52 orders means the daily orders usually vary by about 4–5 orders from the average.

5. Variability

Since the coefficient of variation is only 18.08%, the data is

Moderately Variable

5) Case Study – Comparing Two Machine Learning Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.

Solution:

Model A

2,3,3,4,4,5,5,6

Model B

1,1,2,4,5,7,8,9

1. Mean Error

Model A

Sum

$$32$$

Mean

$$4$$

Model B

Sum

$$37$$

Mean

$$4.625$$

Model	Mean Error
A	4.00

Model	Mean Error
B	4.625

2. Variance

Model A

Squared deviations

$$4 + 1 + 1 + 0 + 0 + 1 + 1 + 4 = 12$$

Variance

$$\frac{12}{8} = 1.5$$

Model B

Variance

$$\frac{63.875}{8} = 7.98$$

Model	Variance
A	1.50
B	7.98

3. Standard Deviation

Model A

$$\sqrt{1.5} = 1.22$$

Model B

$$\sqrt{7.98} = 2.82$$

Model	Standard Deviation
A	1.22
B	2.82

4. More Consistent Model

Model A has the lower standard deviation (1.22), indicating that its prediction errors are less spread out.

Therefore, Model A is more consistent.

5. Is Lower Mean Error Always Better?

No.

A lower mean error alone does not guarantee a better model. Other factors such as variance, consistency, accuracy on unseen data, robustness, and performance metrics (e.g., precision, recall, F1-score, RMSE, or MAE) should also be considered. A model with slightly higher mean error but much more consistent predictions may be preferable in many real-world applications.